Recent Topics in NLP

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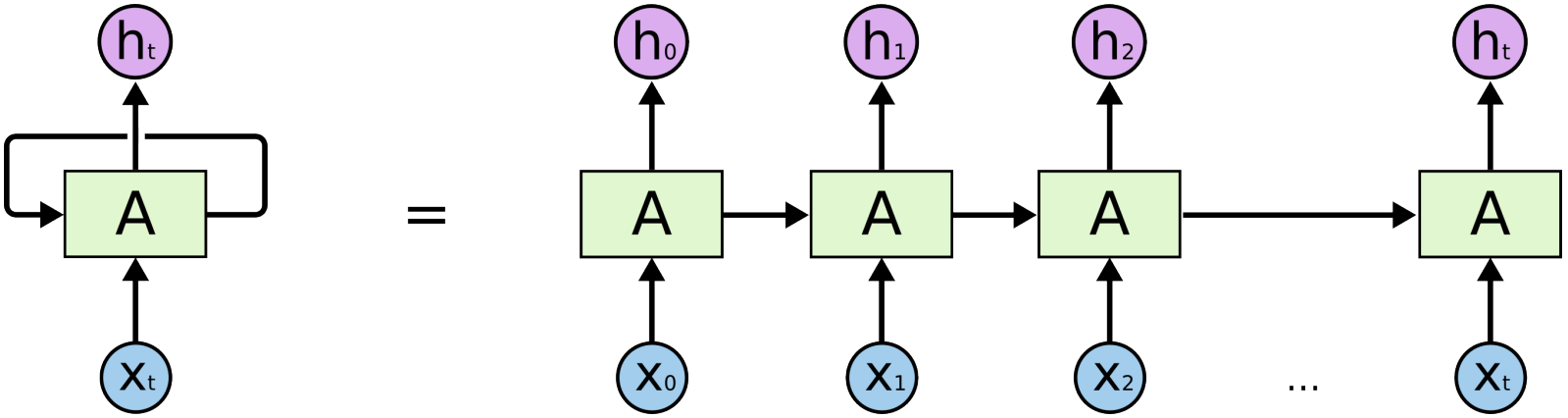
Topics

1. Problems with RNN models
2. Attention Models
3. Last Five Years

The RNN model Unrolled



Basic building block of encoders and decoders for sequence-to-sequence or sequence-to-vector or vector-to-sequence tasks

[https://colah.github.io/posts/2015-08-Understanding-LSTMs/]

Problems with RNNs

- Difficult to learn long-range dependencies (vanishing gradient problem)
- Difficult to parallelize
- Lots of resources needed to train

LSTMs (Long Short-Term Memory) were introduced to fix the first problem.
They contain "forget" gates to forget irrelevant parts of the sequence.

Attention is All You Need

Potential issue with encoder–decoder approach: neural network needs to compress(source sentence info) into fixed-length vector.

What if test data has really long sentences,
$\exists s \in test - corpus\ len(s) > len(t)\ \forall t \in training - corpus$?

Solution: learn to align and translate jointly.

At each t, M generates a word in a translation, it searches for a set of relevant positions in a source sentence

M predicts target word based on the context vectors associated with these source positions and previous generated target words.

Last 5 years

1. Transformer Models (2017) - based on attention idea
2. BERT - Bi-directional transformer based model for generating contextual word embeddings. Uses a masked language model and training on sentence pairs (2018).
3. ELMO (2018)
4. XLNet (2019)